

Trust as a First-Class Routing Objective: Provenance-Robust Resource Selection for AI-Native Data Systems

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Abstract

AI-native data systems answer queries by routing among heterogeneous sources: relational tables, vector indexes, retrieval services, model-backed tools, and third-party or machine-generated stores. Classical resource selection ranks sources by content, but content alone cannot distinguish a curated source from an untrusted duplicate whose text is lexically identical and whose answers may be stale, poisoned, or unverifiable. We formalize provenance-robust resource selection and prove a content-only impossibility bound: with m content-identical impostors, no content-only router can choose the genuine source with probability above $1/(m+1)$ inside the equivalence class. We instantiate trust as a first-class routing objective in TRUSTROUTE, which multiplies a standard CORI relevance signal by an online beta-binomial trust estimate and routes by expected utility under a cost budget. On a reproducible federation of six real information-retrieval collections (23,709 documents; 776 judged queries), TRUSTROUTE remains competitive on a clean federation and is markedly more robust under contamination. With eight evil-twin sources, CORI’s routing accuracy drops from 0.893 to 0.431, whereas TRUSTROUTE retains 0.787, preserving 96% of clean routing accuracy and 96% of clean nDCG@10. At six twins, TRUSTROUTE improves nDCG@10 over CORI by +0.193 and over ReDDE by +0.217, with Holm-corrected paired Wilcoxon tests giving $p \leq 4.9 \times 10^{-59}$ and large effects. Additional experiments show that the posterior recovers graded reliability, tracks non-stationary drift, and

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is stable across a 4×3 hyper-parameter grid.

Keywords: resource selection, federated search, query routing, data provenance, trust calibration, retrieval-augmented generation, AI-native systems

1. Introduction

Modern data systems are no longer closed catalogs of trusted tables. A single analytical or conversational query may be served by a vector index, a lexical search service, a relational store, a model-backed tool, an external knowledge API, or a passage generated on the fly [1–4]. We call such systems *AI-native*: the unit of execution is no longer a stable operator over curated data but a *source* whose reliability, freshness, cost, and—critically—*trustworthiness* vary widely and drift over time.

The component that decides *which source(s) to query* is a *resource-selection* or *query-routing* layer, studied for decades in federated and distributed information retrieval [5–8]. Classical methods rank candidate sources by how well their *content* statistics match the query. This works well when every source is trustworthy. It fails—silently and catastrophically—when some sources are not.

1.1. Why this is now a first-order concern

Provenance-blindness has always been latent in federated retrieval, but three concurrent shifts in how data systems are built have turned it from a corner case into a first-order design problem.

Open data planes and agent tool servers. Retrieval is increasingly brokered across many third-party endpoints registered dynamically rather than curated once. Protocols such as the Model Context Protocol [9] let an agent discover and call external tool and data servers at run time, and “data management for AI” systems treat such heterogeneous sources as first-class [3, 4]. The router now stands in front of endpoints that no one has vetted end-to-end.

Adversarial contamination. Retrieval corpora are demonstrably attackable. Corpus-poisoning attacks craft a handful of adversarial passages that are retrieved for many queries [10], and knowledge-corruption attacks such as PoisonedRAG show that injecting a few malicious documents can flip the answers of a retrieval-augmented generator [11]. A poisoned or hallucinated

31 store can be *lexically indistinguishable* from a genuine one while returning
 32 unverifiable content.

33 *Routing-as-control.* As pipelines add learned and agentic routers [12, 13],
 34 the routing decision—not just the ranking inside a single index—determines
 35 answer quality. If routing is optimized for relevance alone, it inherits exactly
 36 the blind spot that contamination exploits.

37 *The provenance-blindness problem.* Consider a federation that contains, along-
 38 side a curated source, an *untrustworthy duplicate* of it: a scraped copy hosted
 39 by an unverified third party, a stale replica, a poisoned store, or a retrieval
 40 store populated by a hallucinating model. Such a source is, by construction,
 41 *lexically indistinguishable* from the genuine one. Any selector that ranks
 42 sources by content statistics must treat the two identically and will route
 43 to the impostor roughly half the time. Because the impostor’s answers can-
 44 not be trusted—they may be tampered, outdated, or fabricated—this halves
 45 end-to-end answer quality even though the *retrieval* machinery is working
 46 perfectly. Content is simply the wrong signal: the distinguishing property is
 47 *provenance* [14], not text. No amount of better term matching can close a
 48 gap that lives in a dimension term matching cannot see.

49 *Trust as a first-class objective.* We argue that an AI-native router must opti-
 50 mize *trust* as a first-class objective, on equal footing with relevance and cost.
 51 Crucially, trust cannot be read off a source’s content; it must be *learned*
 52 *from observed outcomes*—did routing to this source actually satisfy the in-
 53 formation need? We instantiate this idea in TRUSTROUTE, a router that (i)
 54 computes a standard content-relevance signal, (ii) maintains a per-source *cal-*
 55 *ibrated* trust estimate as a beta–binomial posterior with an optimistic prior
 56 and a lower confidence bound, updated online from routing outcomes, and
 57 (iii) routes by *expected utility* = $\text{rel} \times \text{trust}$ under a cost budget. At cold start
 58 trust is uniform and TRUSTROUTE reduces *exactly* to its content baseline;
 59 as evidence accrues, unreliable sources are multiplicatively suppressed. The
 60 design is deliberately minimal: one beta–binomial pair per source, an $O(1)$
 61 update, and no change to the underlying retriever.

62 *Headline result.* On a fully reproducible federation of six real IR collections,
 63 the gap is stark (Fig. 2). When the federation is diluted with eight evil-twin
 64 sources—so that impostors outnumber genuine sources (8 of 14)—content-
 65 based routing accuracy collapses to 0.431 (CORI) and 0.396 (ReDDE), roughly

66 half of their clean values. TRUSTROUTE holds at 0.787, essentially un-
 67 changed from its clean 0.822. Stated as *robustness retention*—the fraction
 68 of clean accuracy preserved under maximal contamination—TRUSTROUTE
 69 keeps 96% while content baselines keep 48–50% (Table 3). The same pattern
 70 holds for end-to-end nDCG@10 and MAP, is significant after multiplicity
 71 correction, and comes at an average query cost below both content baselines
 72 at TRUSTROUTE’s single-source operating point.

73 *Contributions.*

- 74 • We formalize *provenance-robust resource selection*, give an expected-
 75 utility objective that places trust on equal footing with relevance and
 76 cost, and show, on real data, that content-based selection is structurally
 77 blind to provenance (Sections 3, 4).
- 78 • We prove a *content-only impossibility bound*: any router that observes
 79 only content can place a genuine source first with probability at most
 80 $1/(m+1)$ in a class with m content-identical impostors, no matter how
 81 good the term matcher (Proposition 1).
- 82 • We present TRUSTROUTE, which combines a content-relevance signal
 83 with an online-calibrated beta-binomial trust estimate and routes by
 84 expected utility under a cost budget, reducing to its content baseline
 85 at cold start, with an $O(1)$ trust update and a complexity analysis
 86 (Section 5).
- 87 • We assemble a fully reproducible federated testbed from six real, het-
 88 erogeneous IR collections and evaluate against established baselines
 89 (CORI, ReDDE) with statistical rigor: query-clustered paired Wilcoxon
 90 tests, family-wise Holm–Bonferroni correction, query-bootstrap confi-
 91 dence intervals, and matched-pairs rank-biserial effect sizes over three
 92 seeds (Section 6).
- 93 • We characterize robustness as a *contamination sweep* (0 to 8 twins)
 94 rather than a single point, introduce a retention metric that summarizes
 95 graceful degradation, and report a budget analysis showing where the
 96 trust signal helps and where it does not (Section 7).
- 97 • We *generalize the threat model* beyond exact twins (Section 8): graded/stochastic
 98 poisoning, where the posterior recovers true reliability; non-stationary

99 drift, where the online estimator re-calibrates; and a hyper-parameter
100 sweep that maps the method’s robust operating region and its one hon-
101 estly reported degenerate corner.

- 102 • We report negative and null results candidly, including the modest cost
103 of the mechanism on clean data and the boundary at which content
104 statistics become informative again.

105 *Organization.* Section 2 surveys related work. Section 3 formalizes provenance-
106 robust resource selection and Section 4 the untrusted-source threat model,
107 including the impossibility bound. Section 5 presents TRUSTROUTE. Sec-
108 tions 6 and 7 give the setup and the core results (RQ1–RQ7), Section 8 the
109 generalized threat model (RQ8–RQ10), and Sections 9–10 discuss limitations
110 and conclude.

111 2. Related Work

112 *Resource selection and federated search.* Resource selection ranks collections
113 by their likely usefulness for a query. CORI [5] treats each collection as a
114 “big document” and scores it with an inference-network belief built from per-
115 collection document and collection frequencies. Sample-based methods such
116 as ReDDE [6] estimate the number of relevant documents in each collection
117 by ranking a centralized sample and scaling by collection size. Surveys [7, 8]
118 map the broader space. All of these methods are *content-based*: they are
119 blind to whether a source’s documents can be trusted, which is precisely
120 the failure mode we study. Our contribution is orthogonal to the choice of
121 content signal—we use CORI’s, and add a trust factor that any of these
122 scorers could carry.

123 *Retrieval-augmented generation and learned routing.* RAG pipelines [1, 2]
124 and learned re-rankers [15] have renewed interest in choosing *where* to re-
125 trieve. A recent line of work makes the router itself learned or agentic:
126 agentic RAG systems plan multi-step retrieval and tool use [12], and learned
127 RAG routers select among retrievers or knowledge sources per query [13].
128 These systems optimize routing for relevance, latency, or cost; the trustwor-
129 thiness of the chosen source is typically assumed rather than estimated. Our
130 work supplies that missing objective and shows it is necessary, not merely
131 beneficial: on a contaminated federation, optimizing relevance alone is prov-
132 ably insufficient because relevance cannot separate a twin from its original

133 (Proposition 1). TRUSTROUTE is complementary to a learned router: its
134 calibrated trust factor can be carried by any relevance scorer, learned or
135 hand-built.

136 *Adversarial contamination and corpus poisoning.* A growing body of work
137 shows retrieval corpora are attackable. Corpus-poisoning attacks craft ad-
138 versarial passages that a dense retriever surfaces for a broad set of queries [10];
139 knowledge-corruption attacks such as PoisonedRAG inject a small number
140 of malicious documents to control a RAG system’s output [11]. These de-
141 fenses operate at the *document* level—detecting or filtering individual poi-
142 soned passages. TRUSTROUTE is a *routing-layer* defense, complementary to
143 document-level ones: it asks not “is this passage poisoned?” but “has this
144 *source* historically satisfied information needs?”, and it suppresses a source
145 whose outcomes are unreliable regardless of why. The two layers compose:
146 document filters reduce the per-query poison rate, and the trust layer de-
147 motes sources that remain unreliable in aggregate.

148 *Open data planes, tool servers, and data management for AI.* AI-native data
149 management treats heterogeneous, dynamically registered sources as first-
150 class citizens [3, 4], and agent protocols such as the Model Context Pro-
151 tocol [9] make run-time discovery of external tool and data servers routine.
152 This is exactly the setting in which a source’s provenance cannot be assumed:
153 endpoints appear and disappear, mirrors proliferate, and the router must de-
154 cide whom to trust from behavior, not from a static registry. We position
155 provenance-robust routing as a primitive for this emerging stack.

156 *Provenance, trust, and calibration.* Data provenance characterizes *why* and
157 *where* data originates [14]; we operationalize a downstream consequence—
158 a source whose provenance cannot be verified should not be trusted even
159 when its content matches. Our trust estimator is a beta-binomial poste-
160 rior summarized by a lower confidence bound [16, 17] under an optimistic
161 prior, connecting to optimism-under-uncertainty in bandits [18]. Unlike a
162 bandit that seeks to maximize cumulative reward by exploration, our aim
163 is *calibrated suppression*: we want the posterior to track each source’s true
164 reliability so that routing degrades gracefully, which is why we summarize by
165 a conservative lower bound rather than an optimistic upper one. For evalua-
166 tion we follow best practice for comparing methods across many queries [19]:
167 the Wilcoxon signed-rank test [20], Holm correction [21], and matched-pairs
168 rank-biserial effect sizes [22].

169 3. Problem Formulation

170 **Definition 1** (Federation). *A federation is a set of sources $\mathcal{C} = \{c_1, \dots, c_n\}$.
 171 Each source c_i holds a document collection D_i and answers a query q by re-
 172 turning a ranked list $R_i(q)$ via a local retrieval model (here Okapi BM25 [23]).
 173 Each source has a cost $\text{cost}(c_i)$ and a latency profile.*

174 A router receives a query q and a budget b and must choose an ordered
 175 subset $S(q) \subseteq \mathcal{C}$, $|S(q)| \leq b$, to query; results from the selected sources are
 176 merged into a single ranking $R_S(q)$. We measure *routing accuracy* $R@k$ (is a
 177 source containing relevant documents selected within the top k ?), *reciprocal*
 178 *rank* of the best source, and *end-to-end* retrieval quality (nDCG@10, MAP)
 179 of $R_S(q)$, each as a function of the cost budget b .

180 3.1. An expected-utility objective

181 We make trust an explicit term in the routing objective rather than a
 182 post-hoc filter. Let $\text{rel}(c_i, q) \in [0, 1]$ be a content-relevance score and let
 183 $\text{trust}(c_i) \in [0, 1]$ be a reliability estimate—the probability that routing to c_i
 184 satisfies an information need. We define the *expected utility* of querying c_i
 185 for q as

$$u(c_i, q) \propto \text{rel}(c_i, q) \cdot \text{trust}(c_i) - \lambda \widehat{\text{cost}}(c_i), \quad (1)$$

186 and route to the budget-constrained subset maximizing $\sum_{c \in S} u(c, q)$. The
 187 multiplicative coupling of relevance and trust is the crux: a highly relevant
 188 but untrusted source has *low* utility, so it is demoted exactly when content
 189 scoring would promote it. Setting $\text{trust}(c_i) \equiv 1$ recovers a classical content-
 190 only router, which makes the content-only methods a special case and the
 191 comparison in Section 7 an ablation of the trust term.

192 3.2. Single-home routing

193 Each query’s relevant documents reside in a single *home* source, so se-
 194 lection reduces to routing each query to its home—the standard single-home
 195 federated-search setting [8]. The challenge we introduce is that the federation
 196 may also contain *untrusted* sources whose documents are never relevant de-
 197 spite being lexically attractive (Section 4). A provenance-robust router must
 198 keep routing to genuine homes even as the share of untrusted candidates
 199 grows.

200 4. The Untrusted-Source Threat Model

201 To probe provenance-robustness we inject *evil-twin* sources: exact-content
 202 duplicates of genuine collections, registered under fresh document identifiers
 203 that appear in *no* relevance judgment. A twin models a scraped duplicate,
 204 an unverified mirror, a poisoned store, or a hallucinated one: its documents
 205 look authoritative but are not verifiable, so under the standard pooling as-
 206 sumption (unjudged $\Rightarrow$ non-relevant) they never satisfy an information need.
 207 Because a twin’s content is *identical* to its original, no content-based selec-
 208 tor can rank one above the other—they must break the tie at random, and
 209 route to the impostor with probability that grows with the number of twins.
 210 Only a method that learns from outcomes can route around them. We sweep
 211 the number of injected twins from 0 to 8 and apply identical, seeded ran-
 212 dom tie-breaking to *every* method, so the comparison is scrupulously fair:
 213 TRUSTROUTE is granted no tie-breaking advantage and must *earn* its ro-
 214 bustness from observed outcomes alone.

215 4.1. A taxonomy of untrusted sources

216 Real untrusted sources occupy a spectrum, and the exact twin is its hard-
 217 est, content-blindness-maximizing point. We organize the space along two
 218 axes— *how reliable* the source is and *whether* that reliability is stable.

- 219 • *Exact twin* (reliability 0, stable). Content-identical to a genuine source
 220 but never relevant; the canonical provenance-blindness case (Section 7).
- 221 • *Graded/stochastic mirror* (reliability $p \in (0, 1)$, stable). Returns gen-
 222 uine answers only with probability p —a faithful model of a partially
 223 poisoned store or a stale replica that is sometimes correct (RQ8, Sec-
 224 tion 8).
- 225 • *Non-stationary source* (reliability changes over time). A mirror that is
 226 refreshed, or a fresh source that becomes poisoned, mid-stream (RQ9,
 227 Section 8).
- 228 • *Degraded mirror* (content statistics differ). Corrupted or partial copies
 229 whose lexical fingerprint no longer matches; here content statistics be-
 230 come partly informative and content-based routing can recover, mark-
 231 ing the boundary of the provenance problem (Section 7).

232 4.2. A content-only impossibility bound

233 The failure is not an artifact of CORI; it is a property of any content-only
234 ranking.

235 **Proposition 1** (Content-only upper bound). *Consider an equivalence class*
236 *containing one genuine source and m untrusted twins with identical content*
237 *features. Let a router rank sources by any function f that depends only on*
238 *content. Under fair (uniform) random tie-breaking, the probability that f*
239 *places the genuine source first within the class is at most $1/(m+1)$.*

240 *Proof.* By assumption all $m+1$ sources in the class share identical content
241 features, so f assigns them an identical score. A ranking must order tied
242 items by the tie-breaking rule; under uniform random tie-breaking each of
243 the $m+1$ sources is equally likely to be placed first, so the genuine source is
244 first with probability exactly $1/(m+1)$. Any deterministic tie-break does no
245 better in expectation over the adversary’s choice of which copy is genuine,
246 giving the stated upper bound. $\square$

247 **Remark 1.** *A better term matcher can raise the score of the whole equivalence*
248 *class, improving its rank relative to other classes, but it cannot break*
249 *the provenance tie inside the class: the genuine-first probability is capped at*
250 *$1/(m+1)$ regardless of f . Only non-content evidence—such as the outcome-*
251 *based trust signal of Section 5—can exceed this bound. At $m = 8$ the cap is*
252 *$\frac{1}{9} \approx 0.11$ per contaminated class, which is why content baselines lose roughly*
253 *half their federation-wide accuracy once impostors are plentiful.*

254 5. Trust-Aware Routing

255 5.1. Content relevance signal

For comparability with prior work we use the CORI [5] collection score as the content signal. For query term k and source c_i with document frequency $df_{i,k}$ and total term count cw_i (mean $\overline{cw}$),

$$T_{i,k} = \frac{df_{i,k}}{df_{i,k} + 50 + 150 \cdot cw_i / \overline{cw}}, \quad (2)$$

$$I_k = \frac{\log((|\mathcal{C}| + 0.5)/cf_k)}{\log(|\mathcal{C}| + 1)}, \quad (3)$$

$$\text{rel}(c_i, q) = \frac{1}{|q|} \sum_{k \in q} \left(\beta + (1 - \beta) T_{i,k} I_k \right), \quad (4)$$

256 with cf_k the number of sources containing k and $\beta = 0.4$.

257 *5.2. Calibrated trust*

258 Each source carries a beta-binomial trust posterior $\text{Beta}(\alpha_i, \beta_i)$ with
 259 mean and variance

$$\mu_i = \frac{\alpha_i}{\alpha_i + \beta_i}, \quad \sigma_i^2 = \frac{\alpha_i \beta_i}{(\alpha_i + \beta_i)^2 (\alpha_i + \beta_i + 1)}. \quad (5)$$

260 We rank by a *lower confidence bound*, which penalizes uncertainty:

$$\tau_i = \max(0, \mu_i - z \sigma_i), \quad z = 1. \quad (6)$$

261 The prior is *optimistic*, $(\alpha_0, \beta_0) = (8, 2)$, so an unseen source is trusted by
 262 default and trust is only *demoted* as failures accumulate. This optimism-
 263 under-uncertainty choice [18] is essential: it prevents rarely queried but re-
 264 liable sources from being unfairly penalized, while still suppressing sources
 265 that repeatedly fail. Ranking by the lower bound, rather than the mean, en-
 266 sures that a source which has *not yet* been observed enough to be trusted is
 267 treated cautiously without being condemned. We later show (Section 8) that
 268 the prior and z have a robust operating region and one degenerate corner,
 269 which motivates these defaults.

270 *5.3. Expected-utility routing*

271 The router instantiates Eq. (1) by scoring each source by relevance weighted
 272 by reliability, with a small cost regularizer:

$$s(c_i, q) = \underbrace{\frac{\text{rel}(c_i, q)}{\max_j \text{rel}(c_j, q)}}_{\text{normalized relevance}} \cdot \tau_i - \lambda \widehat{\text{cost}}(c_i), \quad (7)$$

273 with $\lambda = 0.02$ and $\widehat{\text{cost}}$ the min-max-normalized cost. Sources are ranked
 274 by s and the top b are queried. Because τ_i *multiplies* relevance, a high-
 275 relevance but untrusted source is suppressed once its trust falls—exactly
 276 the provenance-blindness gap that content-only scoring cannot close. At
 277 cold start all τ_i are equal and Eq. (7) reduces to the CORI ranking, so
 278 TRUSTROUTE matches its baseline until it has evidence; the only price is
 279 the variance of learning, which we quantify in Section 7.

Algorithm 1: TRUSTROUTE prequential routing over a query stream

```

1. Initialize  $\text{Beta}(\alpha_i, \beta_i) \leftarrow (8, 2)$  for all  $c_i$ 
2. for each query  $q$  in stream do
3.   compute  $\text{rel}(c_i, q)$  for all  $i$  (CORI)
4.    $\tau_i \leftarrow \max(0, \mu_i - z\sigma_i)$ 
5.    $s(c_i, q) \leftarrow \widehat{\text{rel}}(c_i, q) \tau_i - \lambda \widehat{\text{cost}}(c_i)$ 
6.    $S(q) \leftarrow$  top- $b$  sources by  $s$  (random tie-break)
7.   merge  $\{R_c(q) : c \in S(q)\}$ ; emit ranking; record metrics
8.    $c^* \leftarrow \arg \max_i s$ ;  $o \leftarrow \mathbb{I}[R_{c^*}(q) \cap \text{rel}(q) \neq \emptyset]$ 
9.    $\alpha_{c^*} += o$ ;  $\beta_{c^*} += 1 - o$ 
10 end for

```

Figure 1: TRUSTROUTE prequential routing: trust at query t depends only on queries $< t$ (no train/test leakage).

280 *5.4. Online calibration*

281 Trust is updated *prequentially*: each query is scored with the current pos-
 282 terior, then the outcome of routing to the top-1 source c^* is observed and
 283 folded back. The outcome is $o = \mathbb{I}[R_{c^*}(q) \text{ contains a judged-relevant document}]$,
 284 and

$$\alpha_{c^*} += o, \quad \beta_{c^*} += (1 - o). \quad (8)$$

285 Thus the trust used at query t depends only on queries $< t$; there is no
 286 train/test leakage. Algorithm 1 summarizes the procedure.

287 *5.5. Complexity and overhead*

288 TRUSTROUTE adds negligible cost over its content baseline. The state is
 289 two floating-point counters per source, $O(n)$ memory. Per query, computing
 290 rel is identical to CORI; forming s and selecting the top b is an $O(n \log n)$ sort
 291 (in practice $O(n)$ for a single top-1 selection); the posterior update touches a
 292 single source and is $O(1)$. There is *no* training phase, no batch re-estimation,
 293 and no change to the underlying retriever or its index. The trust layer is
 294 therefore a drop-in wrapper around any source-ranking function, and its run-
 295 time overhead is dominated by a single comparison per source—far below the
 296 cost of issuing even one retrieval to a source. We confirm empirically (Sec-
 297 tion 7, RQ7) that at its $b=1$ operating point TRUSTROUTE’s *operational*

Table 1: Federated testbed: six heterogeneous real collections (23,709 documents, 776 judged queries).

Collection	Domain	Docs	Queries	Avg. len	Rel./q
CACM	Computer science	3204	52	53	14.2
MED	Biomedicine	1033	30	154	23.2
NPL	Electrical engineering	11429	93	42	22.4
CRAN	Aeronautics	1400	225	177	7.2
CISI	Library & information science	1460	76	296	41.0
SCIFACT	Scientific claim verification	5183	300	215	1.1
Total	6 domains	23709	776	–	–

cost is in fact lower than that of either content baseline, because the cost regularizer and confident routing let it commit to a single source.

6. Experimental Setup

6.1. Federated testbed

We assemble six openly licensed IR collections spanning six domains into one federation; each is one source. Five are classic collections assembled between the 1960s and 1990s; SciFact [24] is a modern BEIR collection whose queries are natural scientific claims and whose judgments are roughly an order of magnitude sparser (≈ 1.1 relevant documents per query versus 7–41). Documents, queries, and relevance judgments are real; provenance is recorded with the artifact (GitHub mirror commits for the five classic collections, and the release URL with a SHA-256 digest for SciFact). Table 1 summarizes the testbed. Per-source BM25 nDCG@10 ranges from 0.36 (CISI) to 0.69 (MED), consistent with the published literature, validating the retrieval harness.

6.2. Methods

We compare against established federated-IR baselines under identical conditions: *Random* (lower bound), *Search-All* (exhaustive; recall upper bound), *CORI* [5], *ReDDE* [6] (30% centralized sample), and an *Oracle* that routes to the true home source (routing upper bound). TRUSTROUTE is the proposed router. All share the same BM25 sources, the same merge function (within-source min–max normalization), and the same seeded random tie-breaking. To test whether online feedback alone explains the gains, we

also report two online reputation ablations at the six-twin operating point: *CORI+Mean* multiplies CORI by the posterior mean of each source’s success rate, while *CORI+UCB* uses an optimistic upper confidence bound. Both receive the same prequential feedback as TRUSTROUTE.

6.3. Protocol and metrics

Queries from all collections are pooled and streamed in a fixed, seeded order. TRUSTROUTE is evaluated prequentially (Algorithm 1). We report routing accuracy $R@1$, reciprocal rank (MRR), and end-to-end nDCG@10 and MAP at budgets $b \in \{1, 2, 3\}$. All numbers are averaged over seeds $\{7, 11, 13\}$. For confirmatory inference, we first average the three seed replicates within each query, preserving the 776 queries as the independent experimental units. We use two-sided paired Wilcoxon signed-rank tests [20], matched-pairs rank-biserial effect sizes [22], and 10k-resample query-cluster bootstrap confidence intervals. Holm–Bonferroni correction [21] controls the family of all four confirmatory comparisons (two metrics against two baselines).

6.4. Reproducibility

The core artifact is dependency-free; the study uses `numpy/scipy` for vectorized BM25 and statistics and `matplotlib` for figures. The entire core study—data parsing, contamination sweep, multi-seed runs, adversary variants, figures, and $\text{\LaTeX}$ tables—reruns with `python -m trust_aware --federated-study`, and the generalized threat-model study (Section 8) reruns with `examples/run_extended_study.py`. A regression test asserts the headline claim, and an internal consistency check cross-validates the headline constants against the result JSON.

7. Core Results

7.1. RQ1: Is TRUSTROUTE competitive on a clean federation?

On the clean federation (no twins, Table 2a), TRUSTROUTE is strong but, honestly, *not* the best: CORI attains $R@1 = 0.893$ versus TRUSTROUTE’s 0.822 ± 0.014 . This is the expected price of the trust machinery—additional variance from online learning when there are no untrusted sources to defend against. Both far exceed ReDDE (0.785) and Random (0.176). Notably, TRUSTROUTE achieves this at a lower average query cost than either content baseline (0.435 vs. 0.442 for CORI and 0.555 for ReDDE), owing to the cost regularizer in Eq. (7). We highlight this null result deliberately: a method

Table 2: Main results (mean $\pm$ std over seeds {7, 11, 13}, budget $b=1$). (a) clean federation; (b) six evil-twin sources.

Method	R@1	nDCG@10	MAP	MRR
<i>(a) Clean federation (0 untrusted sources)</i>				
Random	0.176 ± 0.003	0.088 ± 0.004	0.071 ± 0.003	0.416 ± 0.001
ReDDE	0.785 ± 0.005	0.403 ± 0.001	0.331 ± 0.001	0.884 ± 0.003
CORI	0.893 ± 0.000	0.461 ± 0.000	0.379 ± 0.000	0.941 ± 0.000
Trust-Aware	0.822 ± 0.014	0.430 ± 0.004	0.365 ± 0.001	0.893 ± 0.009
Oracle	1.000 ± 0.000	0.509 ± 0.000	0.413 ± 0.000	1.000 ± 0.000
<i>(b) Untrusted federation (6 evil-twin sources)</i>				
Random	0.088 ± 0.005	0.045 ± 0.004	0.036 ± 0.004	0.264 ± 0.005
ReDDE	0.405 ± 0.024	0.204 ± 0.017	0.168 ± 0.014	0.658 ± 0.013
CORI	0.451 ± 0.011	0.228 ± 0.006	0.187 ± 0.007	0.698 ± 0.005
Trust-Aware	0.807 ± 0.012	0.421 ± 0.003	0.357 ± 0.001	0.881 ± 0.007
Oracle	1.000 ± 0.000	0.509 ± 0.000	0.413 ± 0.000	1.000 ± 0.000

that defends against contamination should not claim a free lunch on clean data, and TRUSTROUTE does not.

7.2. RQ2: How robust is routing to untrusted sources?

Figure 2 and Table 3 are the central result. We sweep contamination from 0 to 8 injected twins—at eight twins, impostors outnumber the six genuine sources (8 of 14 candidates). Content-based selection holds up to two twins and then *collapses*: CORI’s routing accuracy falls $0.893 \rightarrow 0.843$ at two twins but then plunges to 0.659 at four and 0.431 at eight; ReDDE follows the same cliff. TRUSTROUTE barely moves across the entire sweep (0.822, 0.825, 0.814, 0.807, 0.787), because it learns to distrust the twins from their null outcomes. End-to-end nDCG@10 and MAP (Table 3b,c) tell the same story.

We summarize this with *robustness retention*, the fraction of clean performance preserved at the worst contamination level (eight twins). TRUSTROUTE retains 96% of its clean routing accuracy and 96% of its clean nDCG@10, whereas CORI and ReDDE retain only 48–50%—they lose roughly half of their quality to impostors that they cannot, in principle, see (Proposition 1). At the six-twin operating point (Table 2b), TRUSTROUTE leads CORI by +0.356 R@1 and +0.193 nDCG@10.

Table 3: Contamination sweep (0–8 evil twins; mean over seeds {7, 11, 13}, budget $b=1$). *Retention* is the value at eight twins as a fraction of the clean (zero-twin) value: higher is more provenance-robust.

Method	0	2	4	6	8	Retention
<i>(a) Routing accuracy $R@1$</i>						
Random	0.176	0.119	0.093	0.088	0.077	44%
ReDDE	0.785	0.751	0.585	0.405	0.396	50%
CORI	0.893	0.843	0.659	0.451	0.431	48%
Trust-Aware	0.822	0.825	0.814	0.807	0.787	96%
Oracle	1.000	1.000	1.000	1.000	1.000	100%
<i>(b) End-to-end $nDCG@10$ ($b=1$)</i>						
Random	0.088	0.061	0.046	0.045	0.038	43%
ReDDE	0.403	0.386	0.322	0.204	0.202	50%
CORI	0.461	0.431	0.358	0.228	0.222	48%
Trust-Aware	0.430	0.430	0.426	0.421	0.412	96%
Oracle	0.509	0.509	0.509	0.509	0.509	100%
<i>(c) End-to-end MAP ($b=1$)</i>						
Random	0.071	0.051	0.038	0.036	0.031	43%
ReDDE	0.331	0.319	0.274	0.168	0.166	50%
CORI	0.379	0.357	0.305	0.187	0.182	48%
Trust-Aware	0.365	0.363	0.361	0.357	0.351	96%
Oracle	0.413	0.413	0.413	0.413	0.413	100%

373 The exact-twin setup is the hardest content-blindness case. The artifact
374 also includes two less stylized variants at six untrusted sources: corrupted
375 mirrors with 20% token perturbation and partial mirrors that keep 50% of a
376 source’s documents (Fig. 3, Table 4). The table usefully marks the bound-
377 ary of the claim: when degradation changes content statistics, content-based
378 routing recovers (CORI even edges ahead by 0.06–0.08 $R@1$); the provenance
379 problem is sharpest when the unreliable source remains content-equivalent
380 to the genuine one. TRUSTROUTE is essentially flat across all three vari-
381 ants ($R@1$ 0.802–0.810), trading a small loss on easy, content-distinguishable
382 contamination for robustness on the hard case.

383 7.3. RQ3: Are the gains statistically significant?

384 Table 5 reports query-clustered paired tests at six twins over $N_q = 776$ in-
385 dependent queries. TRUSTROUTE improves $nDCG@10$ over CORI by +0.193
386 (95% CI [0.173, 0.214], $p_{\text{Holm}} = 4.9 \times 10^{-59}$, $r_{\text{rb}} = 0.809$) and over ReDDE
387 by +0.217 (95% CI [0.196, 0.238], $p_{\text{Holm}} = 2.6 \times 10^{-66}$, $r_{\text{rb}} = 0.841$). MAP
388 shows the same picture, with gains of +0.170 over CORI and +0.190 over

Table 4: Adversary-variant sensitivity at six untrusted sources (mean over seeds). Exact twins are content-identical; corrupted and partial variants are degraded mirrors whose content statistics become easier for content-based methods to distinguish.

Variant	CORI R@1	Trust R@1	Δ	CORI nDCG	Trust nDCG	Δ
exact	0.451	0.807	+0.356	0.228	0.421	+0.193
corrupted-20	0.890	0.810	-0.079	0.458	0.423	-0.034
partial-50	0.860	0.802	-0.059	0.438	0.420	-0.018

Table 5: Significance of TRUSTROUTE vs. baselines at six twins. Seed replicates are averaged within query; tests are paired Wilcoxon with family-wise Holm correction, query-bootstrap 95% CIs, and rank-biserial effects.

Metric	vs.	N_q	Δ	95% CI	p_{Holm}	r_{rb} (effect)
nDCG@10	CORI	776	+0.193	[+0.173, +0.214]	4.9e-59	+0.809 (large)
nDCG@10	ReDDE	776	+0.217	[+0.196, +0.238]	2.6e-66	+0.841 (large)
MAP	CORI	776	+0.170	[+0.152, +0.190]	1.5e-66	+0.825 (large)
MAP	ReDDE	776	+0.190	[+0.170, +0.209]	1.5e-74	+0.854 (large)

389 ReDDE ($p_{\text{Holm}} \leq 1.5 \times 10^{-66}$). All four comparisons have large paired effects
390 and reject the null after family-wise correction.

391 7.4. RQ4: How important is the trust estimator?

392 Table 6 and Fig. 4 compare TRUSTROUTE with online reputation abla-
393 tions that receive the same feedback stream. CORI+Mean tests a simple
394 empirical-success multiplier, while CORI+UCB tests optimistic exploration.
395 All online trust variants recover most of the loss suffered by content-only
396 routing; on this testbed CORI+UCB is strongest on quality ($R@1 = 0.841$,
397 nDCG@10= 0.436) and also marginally cheapest (cost 0.428 vs. TRUSTROUTE’s
398 0.433), so TRUSTROUTE is not dominant among feedback-driven methods—
399 we report this rather than hide it. The central effect is therefore first-class
400 learned trust itself rather than any particular posterior summary: once a
401 source’s observed outcomes inform routing, all three variants recover, and the
402 choice between an optimistic upper bound and a conservative lower bound
403 moves quality by less than 0.035 $R@1$ and cost by about 0.005. That the
404 simple content-only CORI sits far below all three ($R@1 = 0.451$) isolates the
405 trust term, not the feedback plumbing, as the source of the gains.

Table 6: Online trust-estimator ablations at six twins (mean $\pm$ std over seeds, budget $b=1$). Bold marks the best value in each column.

Method	R@1	nDCG@10	MAP	cost
CORI	0.451 ± 0.011	0.228 ± 0.006	0.187 ± 0.007	0.441
CORI+Mean	0.832 ± 0.011	0.430 ± 0.001	0.362 ± 0.002	0.429
CORI+UCB	0.841 ± 0.009	0.436 ± 0.003	0.365 ± 0.003	0.428
Trust-Aware	0.807 ± 0.012	0.421 ± 0.003	0.357 ± 0.001	0.433

Table 7: Budget interaction at six twins (end-to-end nDCG@10 and average cost, mean over seeds). TRUSTROUTE leads at every budget; its cheapest budget ($b=1$) is also its best operating point.

Method	$b=1$		$b=2$		$b=3$	
	nDCG@10	cost	nDCG@10	cost	nDCG@10	cost
ReDDE	0.204	0.522	0.293	1.047	0.264	1.665
CORI	0.228	0.441	0.319	0.883	0.281	1.356
Trust-Aware	0.421	0.433	0.371	0.987	0.336	1.380

406 *7.5. RQ5: How does the cost budget interact with trust?*

407 Table 7 reports end-to-end nDCG@10 at budgets $b \in \{1, 2, 3\}$ on the
 408 six-twin federation. TRUSTROUTE leads at every budget, but its most ef-
 409 ficient operating point is $b=1$: routing to the single most-trusted source
 410 yields nDCG@10 0.421 at cost 0.433—better than *any* budget of either
 411 baseline. Increasing the budget actually *lowers* TRUSTROUTE’s nDCG@10
 412 ($0.421 \rightarrow 0.371 \rightarrow 0.336$), because querying additional sources merges in twin
 413 and off-home results that dilute the top of the ranking—a property of the
 414 merge, not of trust. The lesson is sharp: when trust is reliable, spending
 415 budget to hedge across sources is counterproductive; the value of the trust
 416 signal is precisely that it lets a router commit to one source confidently and
 417 cheaply.

418 *7.6. RQ6: Does online calibration work, and what does it learn?*

419 Figure 5 (top) plots mean rolling routing accuracy over all three query
 420 streams at six twins: TRUSTROUTE stabilizes near 0.8 within the first win-
 421 dow and holds it, while CORI and ReDDE remain much lower. The lower
 422 panel shows final trust lower bounds for the representative seed-7 run. Fre-
 423 quently sampled failing twins are demoted well below the optimistic prior; a

424 rarely selected twin remains near the prior, making explicit that the model
 425 changes trust *only* when it receives evidence—never by assumption.

426 7.7. RQ7: What does trust cost?

427 Nothing, in budget terms. TRUSTROUTE carries one beta-binomial pair
 428 per source and an $O(n \log n)$ ranking per query; the trust update is $O(1)$.
 429 Its average query cost at its $b=1$ operating point undercuts both content
 430 baselines (0.433 vs. 0.441 for CORI and 0.522 for ReDDE; Table 7). Two
 431 honest qualifications: at $b=2$ and $b=3$ TRUSTROUTE is dearer than CORI
 432 (0.987 vs. 0.883 and 1.380 vs. 1.356), though those budgets are worse oper-
 433 ating points for it anyway (RQ5); and the CORI+UCB reputation ablation
 434 is marginally cheaper still at $b=1$ (0.428; Table 6). Fig. 6 contrasts clean vs.
 435 untrusted operating points side by side: TRUSTROUTE retains nearly all of
 436 its accuracy where content-based selection sheds half of its own.

437 8. Generalizing the Threat Model

438 The exact-twin model is deliberately the hardest content-blindness case,
 439 but a skeptic may ask whether it is *too* stylized: real impostors are rarely
 440 either perfectly reliable or perfectly useless, and reliability is rarely fixed. We
 441 therefore stress the trust layer along three further axes. All three reuse the
 442 same six-collection testbed, CORI signal, and prequential protocol, and all
 443 reproduce from `examples/run_extended_study.py`.

444 8.1. RQ8: Graded/stochastic poisoning and calibration recovery

445 We replace the all-or-nothing twin with a *graded mirror* that returns a
 446 genuine answer with probability $p \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$ —a faithful model
 447 of a partially poisoned store or a stale replica that is right some of the time.
 448 The question is not only whether TRUSTROUTE routes well, but whether its
 449 posterior *recovers* the true reliability, which is what makes graceful degrada-
 450 tion possible.

451 It does. Table 8 shows that the learned posterior mean tracks the injected
 452 reliability closely across the whole range: the mirror mean-absolute error
 453 between true p and learned $\hat{\mu}$ is 0.068 with Pearson correlation $r = 0.988$,
 454 and genuine sources are correctly learned to be highly reliable ($\hat{\mu} \approx 0.95$).
 455 Figure 7 plots learned trust against true reliability and shows the points lying
 456 close to the diagonal, with the lower confidence bound sitting just below the
 457 mean as the conservative summary used for routing. This directly rebuts the

Table 8: Graded-poisoning calibration (RQ8). Mirrors return genuine answers with probability *True p*; the learned posterior mean $\hat{\mu}$ and lower bound τ recover it (mirror MAE 0.068, Pearson $r = 0.988$). Means over seeds {7, 11, 13}.

Injected source	True p	Realised	Learned $\hat{\mu}$	Learned LCB τ	#obs
mirror	0.00	0.000	0.129 ± 0.000	0.087	52
mirror	0.20	0.222	0.367 ± 0.066	0.292	30
mirror	0.40	0.376	0.417 ± 0.024	0.369	93
mirror	0.60	0.616	0.624 ± 0.036	0.593	225
mirror	0.80	0.798	0.798 ± 0.029	0.756	76
genuine (mean of 6)	0.970	—	0.948	—	—

458 “evil twins are too stylized” objection: the mechanism is not a twin detector,
 459 it is a reliability estimator, and it works for the entire continuum between
 460 genuine and useless.

461 8.2. RQ9: Non-stationary reliability (drift)

462 Reliability is not static: a stale mirror may be refreshed, or a fresh source
 463 may be poisoned. We construct two content-identical mirrors whose reliabil-
 464 ity *changes mid-stream*—one stale-then-refreshed, one fresh-then-poisoned—
 465 with the regime change at the halfway point of the target’s query stream.
 466 Figure 8 plots the trust trajectories. The online estimator re-calibrates after
 467 the change so that the two trajectories *cross*: the source that becomes reli-
 468 able rises and the one that becomes poisoned falls. No static content signal
 469 can produce this behavior, because both mirrors are lexically identical to
 470 their genuine source throughout; only an outcome-driven estimator can fol-
 471 low a moving target. This is the property that makes TRUSTROUTE suitable
 472 for the open, drifting federations of Section 2: trust is continually re-earned,
 473 never assumed.

474 8.3. RQ10: Hyper-parameter sensitivity

475 Finally we map the method’s robustness to its two free knobs: the opti-
 476 mistic prior (α_0, β_0) and the lower-bound width z . Table 9 reports end-to-end
 477 nDCG@10 over a 4×3 grid at six twins. Across every optimistic prior—
 478 all nine cells with $(\alpha_0, \beta_0) \in \{(4, 2), (8, 2), (20, 5)\}$, which is the regime the
 479 method is specified for—it is stable (nDCG@10 $\in [0.376, 0.434]$), and the de-
 480 fault $(8, 2), z=1$ sits comfortably inside that band. Degradation is confined

Table 9: Hyper-parameter sensitivity at six twins (RQ10): end-to-end nDCG@10 over a 4×3 grid of prior (α_0, β_0) and lower-bound width z . Robust across the sensible region; the one degenerate corner (flat prior, aggressive z) motivates the optimistic default. Bold marks the default.

Prior (α_0, β_0)	$z = 0$	$z = 1$	$z = 2$
(1, 1)	0.404	0.331	0.025
(4, 2)	0.421	0.402	0.376
(8, 2)	0.430	0.421	0.406
(20, 5)	0.434	0.432	0.425

to the *flat* prior (1, 1), and there it is graded in z : 0.404 at $z=0$, 0.331 at $z=1$, and a collapse to 0.025 at $z=2$, which we report rather than hide, because a flat prior gives a fresh source high posterior variance and a wide lower bound then drives its trust to zero before any evidence arrives—freezing the router. The failure is interpretable and points directly to the design choice: an *optimistic* prior is what lets a never-yet-observed reliable source be tried at all. With any non-degenerate prior the method is insensitive to z , confirming that the defaults are a safe, principled operating point rather than a tuned sweet spot.

9. Discussion and Limitations

We report limitations candidly. (i) The collections are real and standard but modest in scale; the design is dataset-agnostic and ports directly to large modern federations (e.g., BEIR shards [25]), which is the most important next step for external validity. (ii) We study single-home routing; multi-home queries, where evidence is spread across sources, are a natural extension and would interact with the budget effect in RQ5. (iii) Retrieval is lexical (BM25); the trust mechanism is orthogonal to the base retriever and should compose with neural rankers, since it multiplies whatever relevance score is supplied. (iv) The evil-twin model treats unverifiable duplicates as non-relevant under the standard pooling assumption; the graded, drift, and degraded-mirror studies (Sections 7, 8) show the boundary—when content statistics change, content-based routers can recover, while content-equivalent contamination remains the hard case. (v) The online feedback signal is a proxy for real user or verification feedback; in deployment this outcome would come from downstream answer validation, click signals, or an auditing layer.

506 9.1. Threats to validity

507 *Construct validity.* Our outcome label—“did the routed source return a
508 judged-relevant document?”—is a reasonable but imperfect proxy for trust-
509 worthiness; a source could return relevant-looking but subtly wrong con-
510 tent. The graded-poisoning study (RQ8) partly addresses this by decoupling
511 reliability from content. *Internal validity.* The prequential protocol elimi-
512 nates train/test leakage, and identical seeded tie-breaking is applied to every
513 method, so TRUSTROUTE is granted no structural advantage. *External va-*
514 *lidity.* Six collections in six domains, spanning both classic corpora and a
515 modern BEIR collection, is a genuine federation but not web-scale; the con-
516 tamination sweep, retention metric, and reproduction harness are designed
517 to transfer unchanged to larger testbeds. Absolute metric values are also
518 sensitive to judgment density: SciFact’s ≈ 1.1 relevant documents per query
519 lift every method’s nDCG@10, Oracle included, so absolute values should
520 not be compared across testbed revisions—only differences between meth-
521 ods on a fixed testbed. *Statistical-conclusion validity.* We treat the 776
522 queries as the independent units, average seed replicates within query, and
523 apply family-wise Holm correction across the full comparison family, so the
524 reported significance is conservative.

525 9.2. Ethical considerations and responsible deployment

526 A trust estimator that demotes sources is a governance mechanism, and
527 like any such mechanism it can be misused. Two cautions are worth stating.
528 First, trust here is *behavioral and per-source*, learned from outcomes; it is not
529 a judgment about content quality, authorship, or any protected attribute, and
530 it should not be repurposed as one. Second, an outcome-driven estimator
531 can in principle be gamed by an adversary who supplies genuinely useful
532 answers until trusted and then defects; the drift study (RQ9) shows the
533 estimator re-demotes such a source once it starts failing, but the lag depends
534 on query volume, so safety-critical deployments should pair the routing-layer
535 signal with document-level verification [10, 11] rather than relying on either
536 alone. We see provenance-robust routing as one layer of defense-in-depth,
537 not a complete solution. None of these caveats undermine the core finding:
538 content-based selection is provenance-blind, loses roughly half its quality to
539 impostors it cannot see, and a calibrated, learned trust objective closes the
540 gap at no additional cost.

541 **10. Conclusion**

542 We made the case that trust must be a first-class routing objective in AI-
543 native data systems, and showed why: content-based resource selection can-
544 not see provenance and collapses when untrusted duplicate sources enter the
545 federation—a failure we prove is fundamental, not incidental (Proposition 1).
546 TRUSTROUTE combines a content-relevance signal with an online-calibrated
547 beta-binomial trust estimate and routes by expected utility under a budget.
548 On a fully reproducible federation of six real collections it is competitive
549 on clean data and dramatically more robust to untrusted sources—retaining
550 96% of its clean accuracy under heavy contamination where content baselines
551 retain roughly half—with large, highly significant gains at a cost below both
552 content baselines at its single-source operating point. Beyond exact twins,
553 the learned posterior recovers graded reliability, tracks non-stationary drift,
554 and is robust across its hyper-parameters. We hope the testbed, the contam-
555 ination sweep, the retention metric, and the generalized threat model make
556 provenance-robust routing a standard axis of evaluation for AI-native data
557 systems.

558 **CRedit authorship contribution statement**

559 Jorge Martinez-Gil: Conceptualization, Methodology, Software, Valid-
560 ation, Formal analysis, Investigation, Data curation, Visualization, Writing -
561 original draft, Writing - review and editing.

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569 Data and code availability

570 All datasets, baselines, statistics, figures, and tables in this paper regener-
571 ate from the released artifact at [https://github.com/jorge-martinez-gil/](https://github.com/jorge-martinez-gil/trust-aware)
572 `trust-aware`. The core study reruns with `python -m trust_aware --federated-study`
573 and the generalized threat-model study (Section 8) with `examples/run_extended_study.py`.
574 A regression test asserts the headline robustness claim, and an internal con-
575 sistency check cross-validates the headline constants against the cached result
576 JSON, so every number above is traceable to a fresh run. The bundled IR
577 collections are documented with provenance metadata and per-collection li-
578 cence terms in the artifact: the five classic collections are public domain,
579 while SciFact’s claims are CC BY 4.0 over S2ORC abstracts under ODC-By
580 1.0.

581 Declaration of generative AI and AI-assisted technologies in the 582 manuscript preparation process

583 During the preparation of this work the author used OpenAI Codex to
584 assist with language editing, LaTeX formatting, and submission-compliance
585 checks. After using this tool, the author reviewed and edited the content as
586 needed and takes full responsibility for the content of the published article.

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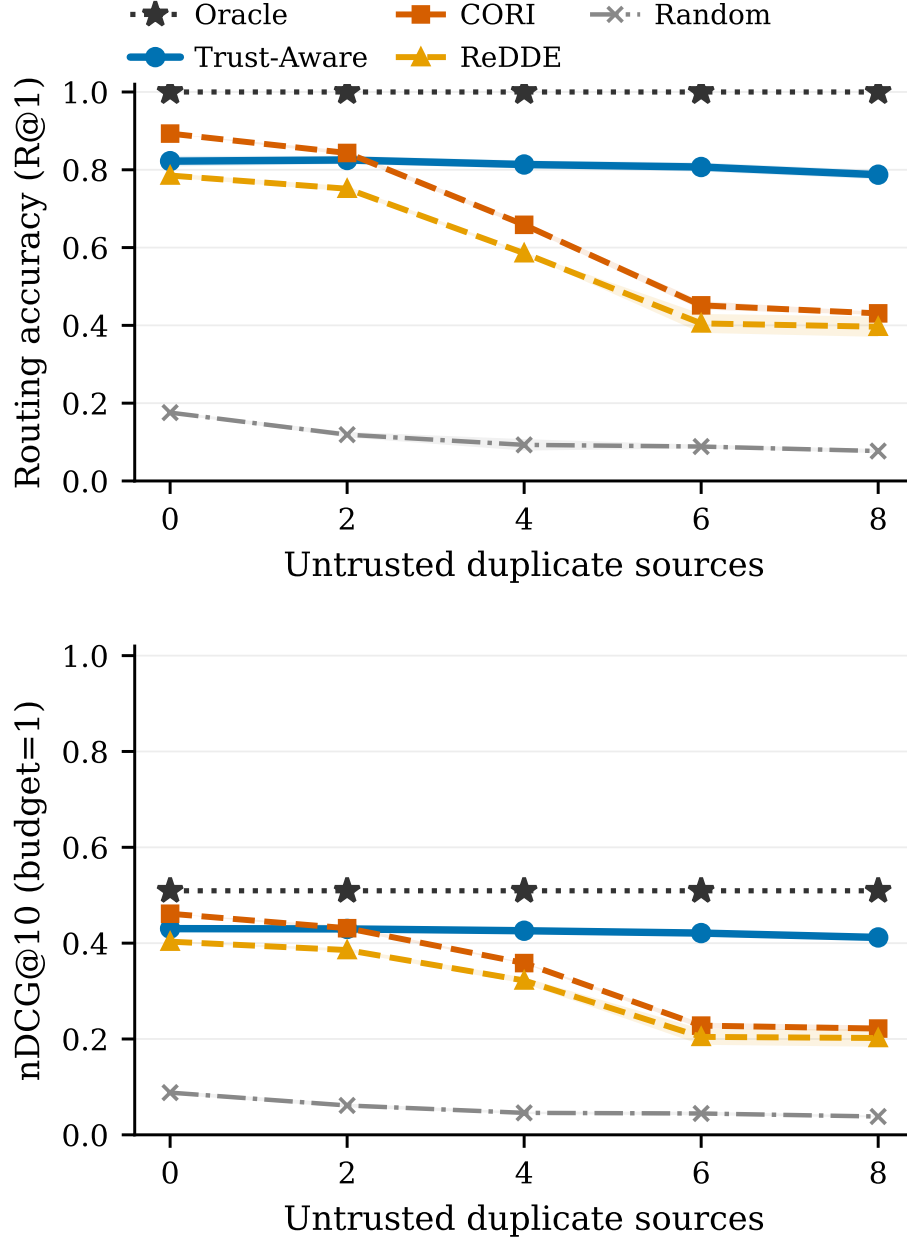


Figure 2: Robustness to untrusted duplicate sources (mean $\pm$ std, 3 seeds). (top) routing accuracy $R@1$; (bottom) end-to-end nDCG@10 at budget 1. Content-based selection (CORI, ReDDE) collapses past two twins; TRUSTROUTE stays flat.

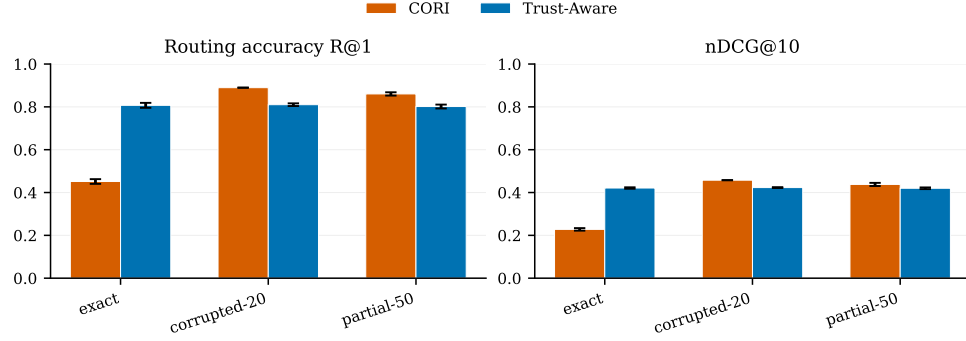


Figure 3: Adversary-variant sensitivity at six untrusted sources. TRUSTROUTE is flat across exact, corrupted-20, and partial-50 mirrors; content-based routing recovers only once the mirror’s content statistics diverge from the genuine source.

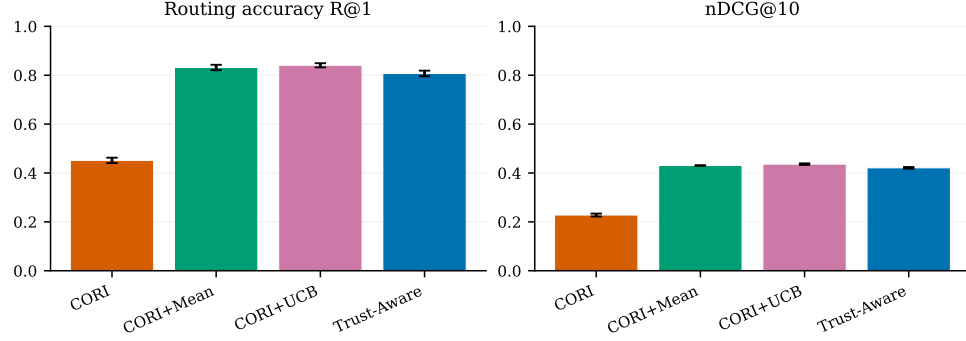


Figure 4: Online reputation ablations at six twins. Calibrated trust (any posterior summary) recovers most of the content-only loss; the choice of summary shifts quality and cost only marginally, with the optimistic CORI+UCB bound slightly ahead of TRUSTROUTE on both.

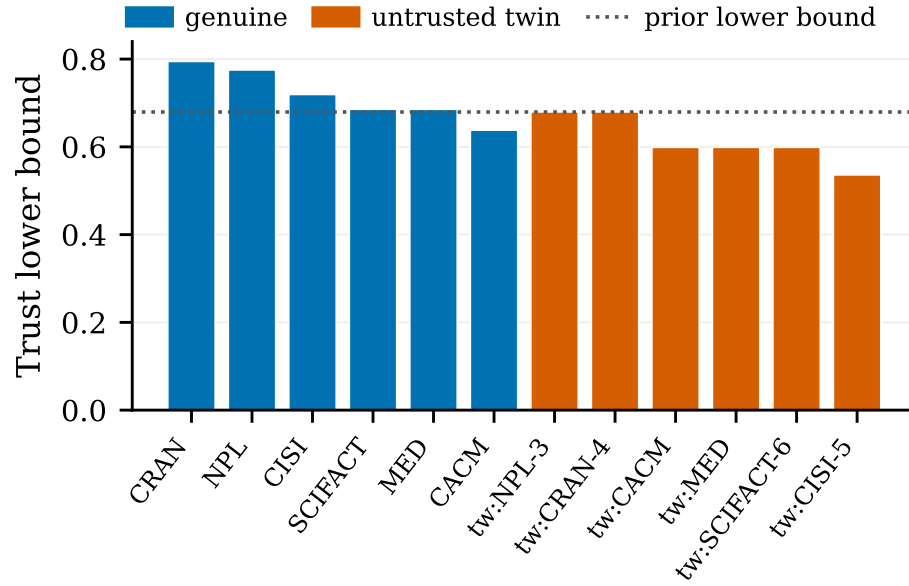
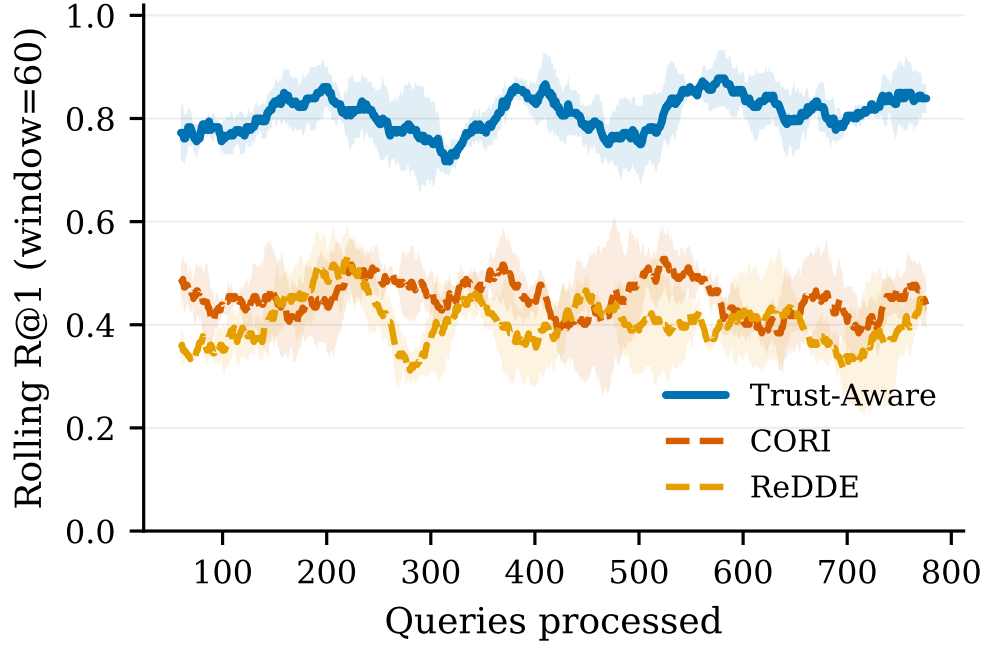


Figure 5: (top) Online calibration at six twins: mean rolling routing accuracy with ± 1 standard-deviation bands across three seeds. (bottom) Final trust lower bounds for seed 7; the dotted line is the optimistic-prior lower bound.

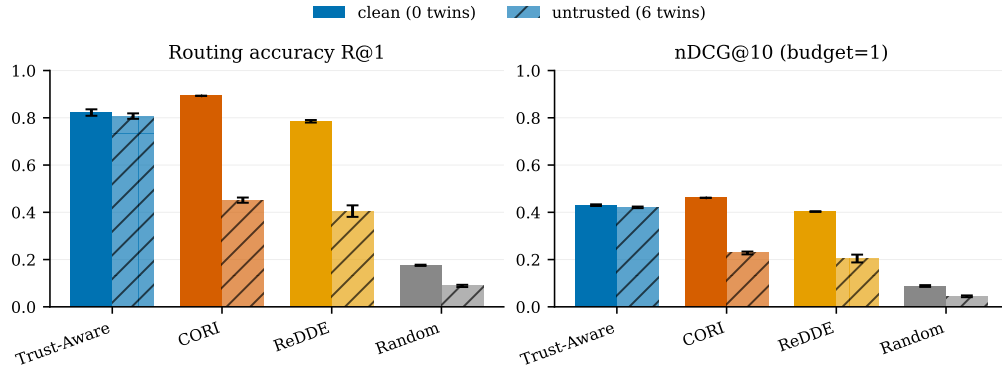


Figure 6: Clean (0 twins) vs. untrusted (6 twins) federations. TRUSTROUTE retains most of its accuracy; content-based selection does not.

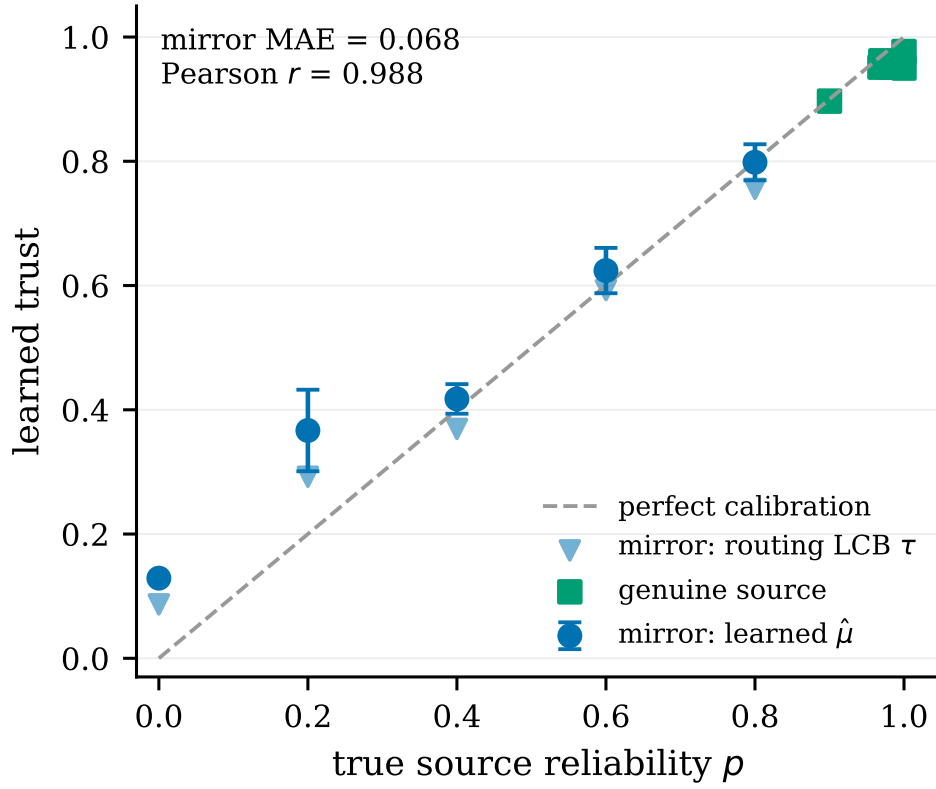


Figure 7: Calibration under graded poisoning (RQ8): learned trust vs. true reliability. Points lie close to the diagonal across the full $[0, 1]$ range; the lower bound (routing summary) sits just below the posterior mean.

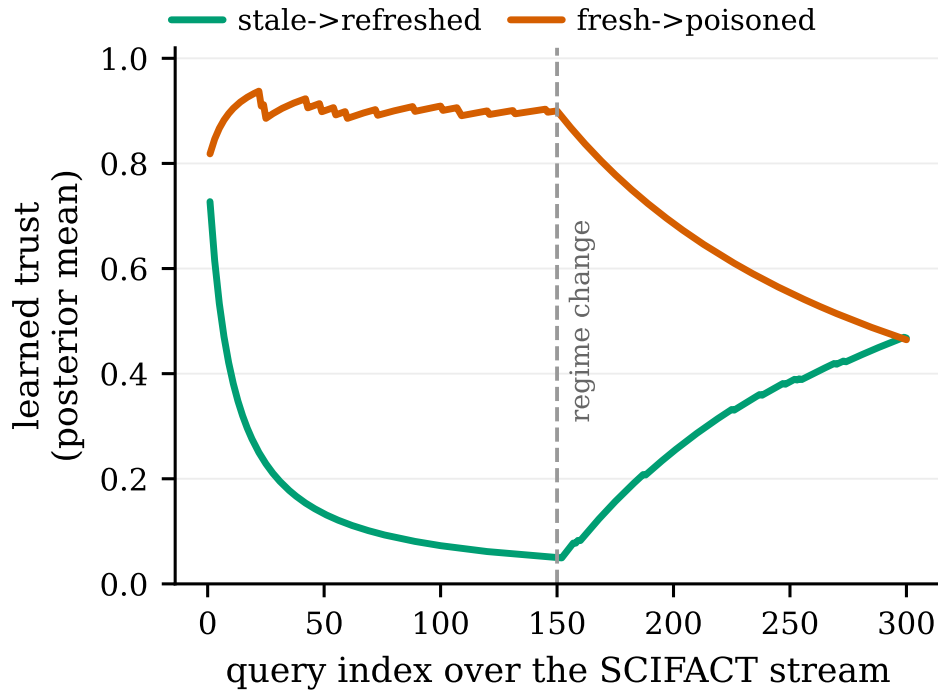


Figure 8: Non-stationary reliability (RQ9). Two content-identical mirrors whose reliability changes at mid-stream (vertical line). The online trust trajectories re-calibrate and cross—behavior no static content signal can exhibit.